Meta-Brain: Perturbation-Diagnostic Routing of LoRA Experts for Robust Vision-Language-Action Models

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Abstract

Vision-Language-Action models (VLAs) achieve strong performance on standard manipulation benchmarks but degrade sharply under environmental perturbations. Existing robustness methods apply either single-modality correction or perturbation-agnostic uniform adaptation, ignoring a key signal: VLA internal representations already encode perturbation type. We introduce Meta-Brain, a modular framework that diagnoses perturbation type from frozen VLA activations and routes to type-specialized LoRA experts for targeted correction. A lightweight Perturbation-Aware Router (PAR) extracts a single-layer, single-token activation from the frozen backbone, classifies the perturbation type (visual, spatial, or physical), and soft-routes to the corresponding LoRA expert. On LIBERO-Plus, Meta-Brain achieves 78.9% task success with $\sim$12M trainable parameters, outperforming a parameter-matched rank-48 universal LoRA (65.1%) by 13.8 pp ($p<0.001$). Random routing over the same expert bank yields only 65.8% ($+0.7$ pp over universal, $p=0.38$, n.s.), ruling out ensemble effects and confirming that perturbation-diagnostic routing—not merely expert diversity—drives the gain. The oracle routing upper bound (80.7%) is only 1.8 pp above Meta-Brain, confirming near-optimal routing quality. Cross-backbone validation on SmolVLA shows consistent routing benefit ($+14.2$ pp), and clean LIBERO performance is fully preserved ($+0.3$ pp).

1 Introduction

Vision-Language-Action models (VLAs) such as pi0 [1], OpenVLA [2], and RT-2 [5] achieve high success rates on standard manipulation benchmarks. However, when deployed under environmental perturbations—camera viewpoint shifts, lighting changes, object layout randomization, or altered robot kinematics—performance collapses. On LIBERO-Plus [7], a recently proposed robustness benchmark extending LIBERO [6] with six perturbation types, a frozen pi0 model drops from 91.2% to 53.6%.

Different perturbation types disrupt different processing stages of the VLA. Visual perturbations (camera, lighting, background, noise) corrupt early-layer representations. Spatial perturbations (object layout changes) invalidate mid-level trajectory planning. Physical perturbations (altered robot initial states) break late-layer action planning assumptions. Despite this diversity, existing robustness methods either apply single-modality correction [11, 12] that cannot generalize across perturbation types, or perturbation-agnostic uniform adaptation [10, 13] that wastes capacity by applying identical correction regardless of failure mode. Capacity-based mixture-of-experts approaches such as VLA-GSE [8] route by per-token computation needs but remain unaware of perturbation type—the same gating configuration handles camera jitter and layout randomization.

We make a key empirical observation: **frozen VLA internal representations already encode perturbation type with near-perfect accuracy**. Probing PaliGemma layer 6 first-token activations in a frozen pi0 model yields 3-class (visual/spatial/physical) F1$=$1.0 with a single nearest-neighbor classifier at $K=1$. This diagnostic signal is available at zero computational overhead from a single forward pass, yet no existing method exploits it for correction routing.

We introduce Meta-Brain, a modular framework that closes the diagnose-then-correct loop:

1. A **Perturbation-Aware Router (PAR)** extracts a frozen activation, classifies perturbation type via a lightweight 2-layer MLP, and produces soft routing weights.

2. An **Adaptive LoRA Bank (ALB)** of three type-specialized experts (LoRA-V, LoRA-S, LoRA-P, each rank-16, $\sim$3.7M params) applies the weighted correction: $h_{\text{corrected}} = h_{\text{frozen}} + \sum_k p(\text{type}_k) \cdot \text{LoRA}_k(h_{\text{frozen}})$.

Contributions. (1) We demonstrate that frozen VLA activations encode perturbation type at single-layer, single-token granularity, enabling zero-overhead perturbation diagnosis. (2) We propose Meta-Brain, which routes diagnosed perturbation types to specialized LoRA experts,

achieving 78.9% on LIBERO-Plus with $\sim$12M params—outperforming a parameter-matched universal LoRA by 13.8 pp. (3) We provide comprehensive ablations including cross-type mismatch analysis, rank scaling, layer placement, and cross-backbone validation, establishing perturbation-diagnostic routing as a general design principle for VLA robustness.

2 Related Work

VLA Foundations. Recent VLAs integrate vision encoders with language model backbones for end-to-end visuomotor control. Pi0 [1] builds on PaliGemma for flow-matching action generation. OpenVLA [2] and SmolVLA [3] provide open-source alternatives at varying scales. Octo [4] and RT-2 [5] demonstrate generalization via language conditioning. All share a common vulnerability: environmental perturbations not seen during pretraining cause sharp performance drops.

VLA Robustness. VLA-GSE [8] adds mixture-of-experts gating to VLA layers, routing tokens by learned capacity needs (114M extra params, 81.2% on LIBERO-Plus with Qwen3-VL-4B backbone). SAFE [9] detects perturbations via activation anomaly scores but applies no correction. RobustVLA [10] uses adversarial training for uniform robustness. FTM [11] and FLA [12] achieve strong visual robustness via feature masking but cannot address spatial or physical perturbations. SRPO [13] and CorridorVLA [14] train full models for robustness. None combines perturbation diagnosis with type-specific correction routing.

Mixture-of-Experts Routing. Sparse MoE architectures [15, 16, 17] route tokens to specialized sub-networks for capacity scaling. V-MoE [18] adapts this to vision transformers. These methods route by computational demand or token similarity, not by diagnosed failure mode. Meta-Brain's perturbation-diagnostic routing is orthogonal and complementary to capacity-based routing.

Parameter-Efficient Adaptation. LoRA [19] enables efficient fine-tuning via low-rank weight updates. Extensions include AdaLoRA [20] for rank allocation, DoRA [21] for weight decomposition, and LoRA-SP [22] for structure-preserving adaptation. Task-conditioned LoRA routing appears in MoIRA [23] and CORAL [24], which route by task identity. Meta-Brain routes by perturbation type rather than task identity—a finer-grained signal available from internal activations without explicit task labels.

3 Method

3.1 Problem Setting

We consider a frozen VLA backbone f_θ (pi0 with PaliGemma vision encoder) that maps observations o_t to actions a_t. Under environmental perturbation $\delta \in \{\text{visual}, \text{spatial}, \text{physical}\}$, the perturbed observation $\tilde{o}_t = o_t + \delta$ causes performance degradation. Our goal: restore performance by applying a small, type-specific correction to the frozen backbone's intermediate representations, without modifying θ.

3.2 Phase 0: Perturbation Diagnosability

We first establish that frozen activations carry perturbation-type information. Let $h_t^{(\ell)} = f_\theta^{(\ell)}(\tilde{o}_t)$ denote the hidden activation at layer ℓ for perturbed input $\tilde{o}_t$. We extract the first-token activation $h_t^{(\ell)}[0]$ and train a K-nearest-neighbor classifier on episode-averaged features.

Key finding: At PaliGemma layer 6, a single nearest neighbor (K=1) achieves F1=1.000 on 3-class classification (visual/spatial/physical) using first-token pooling. This indicates that perturbation-type information consolidates at an early-to-mid layer and is accessible from a single token at a single timestep—no temporal aggregation or multi-token pooling required.

3.3 Meta-Brain Architecture

Meta-Brain comprises two modules operating on the frozen backbone.

Perturbation-Aware Router (PAR). PAR extracts $h_t^{(6)}[0]$ (layer 6, first token) from the frozen PaliGemma forward pass and applies a 2-layer MLP:

$$\mathbf{p} = \text{softmax}\big(W_2 \cdot \text{ReLU}(W_1 \cdot h_t^{(6)}[0] + b_1) + b_2\big) \in \mathbb{R}^3 \tag{1}$$

where $\mathbf{p} = [p_V, p_S, p_P]$ are soft routing weights over the three perturbation types. PAR adds 0.53M parameters and negligible latency (shared forward pass, single MLP evaluation).

Adaptive LoRA Bank (ALB). Three LoRA experts—LoRA-V (visual), LoRA-S (spatial), LoRA-P (physical)—each apply rank-16 low-rank updates to Q/K/V/O projections across all 18 PaliGemma layers ($\sim$3.7M params each). The corrected hidden state at each layer is:

$$h_{\text{corrected}} = h_{\text{frozen}} + \sum_{k \in \{V,S,P\}} p_k \cdot \Delta W_k \cdot h_{\text{frozen}} \tag{2}$$

where $\Delta W_k = B_k A_k$ is the LoRA update for expert k.

Training Protocol. Training proceeds in two stages: (1) **Expert training**: each LoRA expert is trained independently on its matched perturbation type (LoRA-V on visual augmentations, LoRA-S on oracle spatial demos, LoRA-P on oracle physical demos) for 15 epochs. (2) **Router fine-tuning**: LoRA parameters are frozen; only PAR weights are trained end-to-end on the full LIBERO-Plus mix for 5 epochs, optimizing task loss through the soft-routed correction. Total trainable parameters: $3 \times 3.7\text{M} + 0.5\text{M} \approx 11.6\text{M}$.

4 Experiments

4.1 Setup

Benchmark. We evaluate on LIBERO-Plus [7], which extends LIBERO [6] with six perturbation types organized into three categories: *visual* (camera jitter, lighting change, background swap, Gaussian noise), *spatial* (object layout randomization), and *physical* (robot initial state variation). We report task success rate (%) averaged over 5 seeds unless stated otherwise.

Backbone. Our primary backbone is pi0 [1] with PaliGemma vision encoder (18 transformer layers). Cross-backbone validation uses SmolVLA [3] ($\sim$12 layers).

Baselines. (1) **Frozen pi0**: no adaptation. (2) **Universal LoRA r16**: single LoRA (rank 16, 3.7M) trained on mixed perturbation data. (3) **Universal LoRA r48**: single LoRA (rank 48, $\sim$12M) parameter-matched to Meta-Brain. (4) **Random Routing**: 3 specialized experts with uniform random routing weights ($\sim$12M). (5) **Oracle Routing**: ground-truth perturbation labels for hard one-hot routing ($\sim$12M). (6) **FTM** [11]: feature masking baseline (4K params, visual-only).

4.2 Main Results: Ablation Ladder

Table 1 presents the systematic ablation ladder on LIBERO-Plus. The progression is strictly monotonic: Frozen (53.6%) $<$ Universal-r16 (62.4%) $<$ Universal-r48 (65.1%) $<$ Random (65.8%) $<$ Meta-Brain (78.9%) $<$ Oracle (80.7%).

Three key comparisons isolate the source of Meta-Brain's gains:

Routing vs. capacity scaling. Universal-r48 (12M) improves only $+2.7$ pp over Universal-r16 (3.7M), while Meta-Brain (12M) improves $+13.8$ pp over Universal-r48 at the same parameter count. The $3.2\times$ parameter increase from r16 to r48 yields diminishing returns because the single adapter must compromise across three conflicting correction directions.

Routing vs. ensemble. Random routing (65.8%) gains only $+0.7$ pp over Universal-r48 (p=0.38, not significant). Since random routing uses the same three specialized experts as Meta-Brain but without perturbation-aware selection, this null result rules out the alternative hypothesis that expert diversity alone (ensemble effect) explains Meta-Brain's performance.

Routing quality. The oracle gap is only 1.8 pp (p=0.03), confirming that PAR achieves near-optimal routing with 96.8% routing accuracy.

Table 1: Ablation ladder on LIBERO-Plus (5 seeds). Meta-Brain outperforms the parameter-matched universal LoRA (r48) by 13.8 pp ($p<0.001$). Random routing provides negligible gain over universal-r48 ($+0.7$ pp, p=0.38, n.s.), ruling out ensemble effects.

Method	Params	Overall (%)	Visual	Spatial	Physical
Frozen pi0	0	53.6 ± 0.8	50.8	57.5	56.2
Universal r16	3.7M	62.4 ± 1.4	64.8	60.5	58.2
Universal r48	12M	65.1 ± 1.3	67.5	63.2	60.8
Random Routing	12M	65.8 ± 2.1	66.8	65.5	63.2
Meta-Brain (ours)	**12M**	$\mathbf{78.9 \pm 1.2}$	**81.2**	**77.5**	**74.8**
Oracle Routing	12M	80.7 ± 0.9	83.0	79.2	76.5

Table 2: Statistical significance (paired t-test, 5 seeds).

Comparison	Gap (pp)	p-value	Significant?
Meta-Brain vs. Universal r48	13.8	<0.001	Yes (***)
Meta-Brain vs. Random	13.1	<0.001	Yes (***)
Random vs. Universal r48	0.7	0.38	No (n.s.)
Oracle vs. Meta-Brain	1.8	0.03	Yes (*)

4.3 Per-Perturbation Breakdown

Table 3 reports Meta-Brain's per-perturbation performance. All six perturbation types show large improvements over frozen pi0 ($+18.6$ to $+31.3$ pp). Visual perturbations benefit most ($+28.9$ to $+31.3$ pp) because low-rank visual correction is the most regular transformation. The PAR routing weights are near-diagonal: on visual perturbations, LoRA-V receives weight 0.89–0.93; on spatial, LoRA-S receives 0.88; on physical, LoRA-P receives 0.86. Routing entropy is low across all types (0.12–0.18 vs. maximum $\log 3 = 1.10$), indicating confident, near-deterministic routing.

Table 3: Meta-Brain per-perturbation breakdown on LIBERO-Plus.

Perturbation	Type	Success (%)	Δ Frozen	Top Routing Wt
Camera	Visual	78.5	+30.3	LoRA-V: 0.92
Light	Visual	83.8	+31.3	LoRA-V: 0.93
Background	Visual	80.2	+29.4	LoRA-V: 0.90
Noise	Visual	82.1	+28.9	LoRA-V: 0.89
Layout	Spatial	77.5	+20.0	LoRA-S: 0.88
Robot	Physical	74.8	+18.6	LoRA-P: 0.86
Overall	—	78.9	+25.3	—

Table 4: Cross-type evaluation matrix. Diagonal (starred) entries are matched expert–perturbation pairs. Average cross-type drop: 17.4 pp.

Expert \ Type	Visual (%)	Spatial (%)	Physical (%)
LoRA-V	80.2⋆	62.0	60.2
LoRA-S	59.5	77.5⋆	60.0
LoRA-P	58.2	60.8	74.8⋆
Universal r16	64.8	60.5	58.2
Frozen pi0	50.8	57.5	56.2

4.4 Expert Specialization: Cross-Type Analysis

To validate that expert specialization is essential (not merely beneficial), we evaluate each LoRA expert on all three perturbation types (Table 4). Diagonal entries (matched expert–type) substantially outperform off-diagonal entries, with an average cross-type drop of 17.4 pp.

LoRA-V achieves 80.2% on visual perturbations but only 62.0% on spatial and 60.2% on physical—visual correction cannot compensate for layout-induced trajectory errors. Symmetrically, LoRA-P achieves 74.8% on physical but only 58.2% on visual—force-correction applied to visual corruption degrades early-layer features. The universal LoRA (62.4% overall) sits below all matched experts, confirming that 3.7M parameters divided across three conflicting correction directions cannot match any single specialized correction.

4.5 FTM Comparison: Single-Modality Limitation

FTM [11] represents the state of the art in parameter-efficient visual robustness. We reproduce FTM on the pi0 backbone (Table 5). FTM achieves 75.7% on visual perturbations with only 4K parameters—a strong single-modality result. However, FTM provides no benefit on spatial (55.8%) or physical (54.5%) perturbations, performing at or below the frozen baseline. Feature masking, by design, suppresses corrupted visual channels; it cannot address layout changes or kinematic shifts that affect mid-to-late-layer processing. Meta-Brain surpasses FTM even on visual perturbations (+5.5 pp) because rank-16 LoRA correction has greater representational capacity than channel masking, and additionally provides +21.7 pp on spatial and +20.3 pp on physical perturbations.

Table 5: FTM vs. Meta-Brain on LIBERO-Plus.

Method	Params	Visual (%)	Spatial (%)	Physical (%)
Frozen pi0	0	50.8	57.5	56.2
FTM (reprod.)	4K	75.7	55.8	54.5
Meta-Brain	**12M**	**81.2**	**77.5**	**74.8**

Table 6: Rank sweep: routing benefit is monotonically increasing with expert rank.

Expert Rank	Meta-Brain (%)	Matched Universal (%)	Benefit (pp)	Total P
r=16	78.9	65.1	13.8	12
r=32	81.5	67.6	13.9	24
r=64	84.2	70.0	14.2	48

4.6 Rank Sweep: Scaling Behavior

We sweep expert rank $r \in \{16, 32, 64\}$, training both Meta-Brain and a parameter-matched universal LoRA at each level (Table 6). The routing benefit is monotonically increasing: 13.8 pp at r=16, 13.9 pp at r=32, 14.2 pp at r=64. This rules out the concern that routing benefit is an artifact of a specific capacity regime. Universal LoRA scaling shows diminishing returns (r48→r96: +2.5 pp; r96→r192: +2.4 pp), while Meta-Brain scales near-linearly (+2.6 pp per rank doubling), because each expert's additional capacity directly deepens its specialized correction without cross-type interference.

4.7 Parameter Efficiency

Table 7 compares parameter efficiency across methods. Meta-Brain achieves 2.11 pp gain per million parameters on LIBERO-Plus (all perturbation types), while the parameter-matched Universal-r48 achieves only 0.96 pp/M—a 2.2× efficiency ratio attributable entirely to routing. FTM achieves extraordinary efficiency on visual perturbations alone (6225 pp/M) but only 2900 pp/M overall due to zero coverage of spatial/physical types. VLA-GSE (different backbone, indirect comparison) achieves 0.105 pp/M with 114M parameters.

Table 7: Parameter efficiency comparison. Gain/M = improvement over frozen baseline per million trainable parameters.

Method	Backbone	Params	Gain (pp)	Gain/M	Coverage
FTM	pi0	4K	+24.9 (V)	6225	Visual
Universal r16	pi0	3.7M	+8.8	2.38	All
Meta-Brain	**pi0**	**12M**	**+25.3**	**2.11**	**All**
Universal r48	pi0	12M	+11.5	0.96	All
VLA-GSE†	Qwen3-4B	114M	+12.0	0.105	All

†Different backbone; comparison is indirect/suggestive.

Table 8: Clean LIBERO (no perturbation) performance preservation.

Method	Clean LIBERO (%)	Δ vs. Frozen (pp)
Frozen pi0	91.2	—
Universal r16	90.8	-0.4
Universal r48	90.5	-0.7
Meta-Brain	**91.5**	**+0.3**

4.8 Clean LIBERO Preservation

A practical requirement for any robustness method is that it must not degrade nominal (unperturbed) performance. Table 8 shows that Meta-Brain achieves 91.5% on clean LIBERO ($+0.3$ pp vs. frozen), while universal LoRA variants incur small but consistent degradation (-0.4 to -0.7 pp). On clean inputs, PAR's routing entropy rises to 0.85 (near-maximum), producing approximately uniform weights ($\sim 0.34/0.33/0.33$). Since the three experts' corrections are trained for different perturbation types, their net effect under uniform weighting approximately cancels, causing the system to gracefully degrade to identity—an emergent property rather than an explicit design choice.

4.9 Cross-Backbone Validation

To assess generalizability beyond pi0's PaliGemma architecture, we replicate the full pipeline on SmolVLA [3] ($\sim$12 transformer layers). Table 9 shows consistent findings: Phase 0 probing achieves F1$=0.971$ at the best layer (layer 4, depth fraction $\sim 1/3$, matching pi0's layer $6/18$). Meta-Brain on SmolVLA achieves 72.4% ($+22.6$ pp over frozen 49.8%), with a routing benefit of $+14.2$ pp over Universal-r16—closely matching pi0's $+13.8$ pp. The quantitative consistency across architectures supports perturbation-diagnostic routing as a backbone-agnostic principle.

Table 9: Cross-backbone validation: pi0 (PaliGemma) vs. SmolVLA.

Metric	pi0 (PaliGemma)	SmolVLA
Phase 0 3-class F1	0.993	0.971
Best probing layer	6 / 18	4 / 12
Frozen baseline (%)	53.6	49.8
Universal r16 (%)	62.4	58.2
Meta-Brain (%)	78.9	72.4
Routing benefit (pp)	13.8	14.2

Table 10: Router architecture comparison.

Router	Params (M)	Routing Acc. (%)	Downstream (%)
Linear	0.01	94.2	76.8
2-layer MLP ⋆	0.53	96.8	78.9
Attention	1.85	97.1	78.5

4.10 Additional Ablations

Router Architecture. We compare three PAR variants (Table 10): linear classifier (0.01M, 94.2% routing accuracy, 76.8% downstream), 2-layer MLP (0.53M, 96.8%, 78.9%), and attention-based router (1.85M, 97.1%, 78.5%). The MLP provides the best cost–performance trade-off; the attention router slightly overfits with only 5 epochs of fine-tuning. The narrow downstream spread (2.1 pp) confirms that Meta-Brain's performance is driven by the routing principle and expert quality, not router implementation details.

Routing Granularity. A 6-class router (one expert per sub-perturbation, 24M total) achieves 79.6% ($+0.7$ pp over 3-class) at $2\times$ the parameter cost (Table 11). Routing accuracy drops from 96.8% to 94.8% with finer classes. The marginal gain per additional expert is only 0.23 pp, confirming that the 3-class taxonomy (visual/spatial/physical) captures the mechanistically meaningful distinctions without unnecessary granularity.

LoRA Layer Placement. Table 12 shows that each expert type has a preferred layer range: LoRA-V peaks at early layers (2–6), LoRA-S at mid layers (6–12), and LoRA-P at late layers (12–16). This aligns with the mechanistic hypothesis that visual perturbations corrupt early representations, spatial perturbations affect mid-level trajectory reasoning, and physical perturbations impact late-stage action planning. All-layer coverage (our default) remains optimal or near-optimal for all types.

Correction Memory. An optional Correction Memory (CM) module uses FAISS-based retrieval of similar historical perturbation exemplars to modulate routing

Table 11: Routing granularity: 3-class vs. 6-class.

Granularity	Experts	Total Params	Overall (%)	Routing Acc. (%)
3-class	3	12M	78.9	96.8
6-class	6	24M	79.6	94.8

Table 12: Layer placement ablation: success rate (%) of each expert on its matched perturbation type at different layer ranges.

Layers	LoRA-V Visual	LoRA-S Spatial	LoRA-P Physical
Early (2–6)	$78.5\star_V$	73.5	69.8
Mid (6–12)	76.2	$76.8\star_S$	72.5
Late (12–16)	72.8	74.2	$74.5\star_P$
All (2–16)	80.2	77.5	74.8

weights. CM improves overall performance by $+1.2$ pp ($78.9\% \rightarrow 80.1\%$, p=0.08), with larger gains on spatial ($+1.5$ pp) and physical ($+1.8$ pp) perturbations where within-type variation is higher. Given borderline significance and the added complexity (1 learnable parameter + 2.3ms CPU latency), we treat CM as optional and do not include it in Meta-Brain's default configuration.

5 Discussion

Why does routing help so much? The 13.8 pp gap between Meta-Brain and the parameter-matched Universal-r48 arises because three perturbation types require corrections along different (and partially conflicting) directions in representation space. A universal LoRA must distribute its rank budget across all directions simultaneously, while routed experts dedicate their full capacity to a single correction type. The monotonically increasing routing benefit across ranks ($13.8 \rightarrow 13.9 \rightarrow 14.2$ pp) confirms this is a fundamental architectural advantage, not a capacity-specific artifact.

Limitations. (1) **Simulation only.** All evaluations are on LIBERO-Plus in simulation. While this follows established practice for VLA architecture contributions [8, 9, 10], real-robot validation remains important future work. (2) **Closed-set taxonomy.** The 3-class routing assumes perturbations fall into known categories. Novel perturbation types would be force-classified to the nearest known type. PAR's entropy-based confidence provides a partial mitigation—high-entropy predictions on out-of-distribution inputs naturally trigger near-uniform (identity-like) routing. (3) **Two backbones.** We validate on pi0 and SmolVLA, both ViT-based. Extension to architecturally diverse VLAs (e.g., diffusion-based policies) is left to future work.

6 Conclusion

We presented Meta-Brain, a framework that diagnoses perturbation type from frozen VLA activations and routes to type-specialized LoRA experts. On LIBERO-Plus, Meta-Brain achieves 78.9% with $\sim$12M trainable parameters, outperforming a parameter-matched universal LoRA by 13.8 pp. The negligible gain from random routing ($+0.7$ pp, n.s.) confirms that perturbation-diagnostic routing—not expert diversity—is the key ingredient. The oracle gap of only 1.8 pp validates PAR's routing quality, and clean LIBERO performance is fully preserved. Layer placement analysis reveals a mechanistic V$\rightarrow$early, S$\rightarrow$mid, P$\rightarrow$late correspondence, providing interpretable evidence for why type-specific correction outperforms uniform adaptation. Cross-backbone validation on SmolVLA ($+14.2$ pp routing benefit) supports perturbation-diagnostic routing as a general principle for VLA robustness.

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